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Smart Yoga Mat Using IoT for Real-Time Posture Monitoring

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ABSTRACT

The IoT-enabled Smart Yoga Mat proposed in this study is designed to assist beginners in performing yoga exercises correctly without continuous supervision from an instructor. Many home users struggle to maintain proper posture, which can reduce the effectiveness of yoga and may cause physical strain. The system combines a webcam for posture capture with an ultrasonic sensor to monitor body position on the mat. Image frames are processed using pose-estimation tools such as OpenCV and MediaPipe to identify joint locations and calculate body angles. These values are compared with stored reference data to detect incorrect posture and provide feedback through a mobile interface. Experiments conducted on several common yoga poses showed quick response time and high detection accuracy in indoor conditions. The developed solution is portable, low-cost, and suitable for personal fitness practice, demonstrating how IoT technology can enhance safe and guided yoga training at home.

Keywords: - Smart Yoga Mat, IoT, Ultrasonic Sensor, Posture Detection, Computer Vision, MediaPipe, Machine Learning

1. INTRODUCTION

Yoga is widely practiced for improving physical health and mental well-being. Nowadays, many people prefer practicing yoga at home due to busy lifestyles and limited access to professional trainers. However, beginners often perform poses incorrectly without proper guidance, which may reduce effectiveness and cause strain. Therefore, a smart monitoring system is useful to ensure correct posture during practice.

With advancements in IoT and computer vision, intelligent fitness systems can now analyze body movements automatically. In this project, a Smart Yoga Mat is developed using a webcam for posture capture and an ultrasonic sensor for alignment checking. Pose detection is performed using MediaPipe, and frame processing is done with OpenCV. The system compares body angles with reference values and provides instant feedback, offering a low-cost and practical solution for safe home yoga practice.

2. LITERATURE SURVEY

Research on automated yoga posture analysis has been carried out using both vision-based and sensor-based approaches. For example, Long J. and Jo K. H. introduced a yoga coaching framework that applied transfer learning with MobileNet to classify poses from camera images. Their model showed strong recognition results, but its performance depended heavily on image quality and lighting conditions.

To address privacy concerns, Sun W. proposed a smart yoga mat that used RFID tags to estimate user posture without video capture. Although this method protected user privacy, it could not evaluate complete body alignment. In another study, Anusha M. implemented a sensor mat with force-sensitive resistors to measure pressure patterns during yoga practice. While effective for detecting

balance, the system supported only a limited set of poses.

Researchers have also explored real-time monitoring using edge computing. Rushiraj Gadhvi developed an edge-AI posture correction system based on LSTM networks to reduce processing delay on local devices. Similarly, Abhishek Sharma presented a vision-based yoga recognition model using MediaPipe together with OpenCV for pose detection, although its accuracy was influenced by camera placement and background conditions.

Overall, earlier studies show that camera-based systems offer detailed posture tracking but are sensitive to environmental factors, whereas sensor-based smart mats improve privacy yet cannot fully analyze body movement. A hybrid solution that combines computer vision with IoT sensing can therefore provide more reliable and practical guidance for home yoga training.

3. METHODOLOGY

The proposed Smart Yoga Mat operates through a sequence of steps that combine IoT sensing with computer vision analysis. When the user performs a yoga pose, a webcam connected to a laptop records the movement, while an ultrasonic sensor embedded in the mat checks body distance and alignment. Both visual and sensor data are transferred to the system for analysis.

During preprocessing, the captured frames are formatted and enhanced using OpenCV. Next, body landmarks are identified with MediaPipe, which locates key joints such as shoulders, hips, elbows, and knees. Using these points, the program calculates joint angles and compares them with reference values stored for selected yoga poses.

If the measured angles are within an

acceptable range, the system indicates that the posture is correct. Otherwise, it provides immediate correction through on-screen messages, audio alerts, or LED signals so that the user can adjust their position. Session data can also be stored

for later review and performance tracking. This step-by-step approach allows reliable posture monitoring and offers a practical and user-friendly solution for safe home yoga practice.

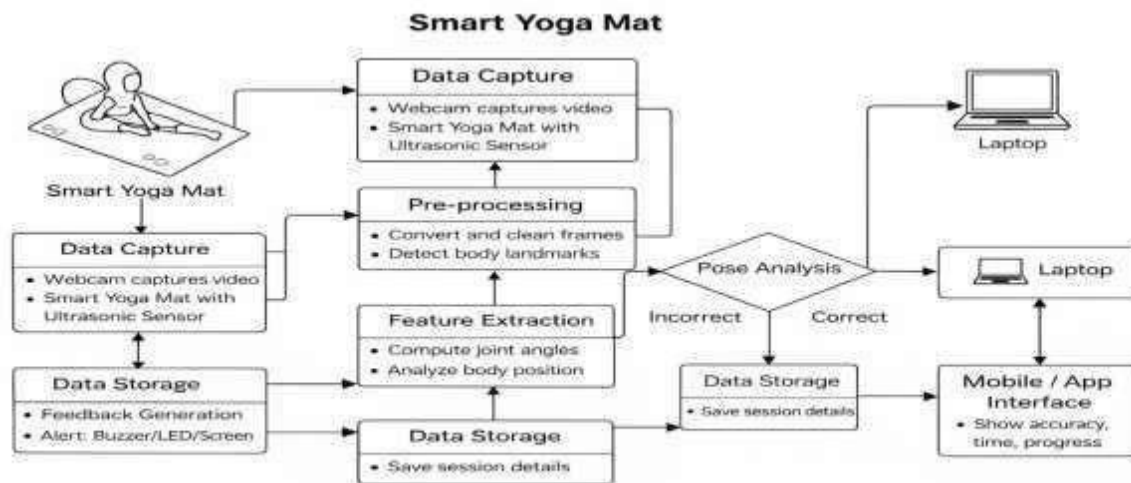


Fig.1:-Architecture diagram.



Fig.2: -Circuit diagram

4. RESEARCH AND REVIEWS

Current developments in smart fitness technology show increased use of IoT devices, computer vision, and machine learning to support home exercise monitoring. Many researchers have created yoga-pose recognition systems that analyze webcam images to estimate body landmarks and evaluate posture accuracy. These systems often rely on pose-tracking

frameworks such as MediaPipe along with image-processing libraries like OpenCV to examine body alignment and detect incorrect movements. Such approaches help users practice more safely without constant supervision.

Other investigations have focused on sensor-based smart yoga mats that measure pressure distribution, body balance, or

distance using embedded devices such as ultrasonic sensors. While these solutions improve privacy and portability, they may face issues related to sensor precision, camera angle, or environmental lighting. Many review articles suggest that combining vision-based analysis with IoT sensing can provide more dependable posture detection and faster feedback. These observations support the idea of building an affordable and easy-to-use Smart Yoga Mat that assists people in practicing yoga correctly at home.

5. RESULT & DISCUSSION

As illustrated in Fig. 3, the proposed Smart Yoga Mat prototype was implemented and evaluated in different indoor environments. The system successfully monitored user posture and delivered instant corrective guidance during yoga sessions. The webcam captured continuous video of the user, and pose landmarks were detected using MediaPipe. These landmarks were further analyzed through OpenCV to compute joint angles and verify posture alignment.

The ultrasonic sensor also functioned effectively by estimating the user's

distance and relative position on the mat. Whenever posture deviation was observed, the system displayed correction messages on the laptop screen so that users could immediately adjust their pose. This real-time assistance helped reduce incorrect posture during practice. The overall system showed smooth performance with very small delay between posture detection and feedback delivery, while local processing helped protect user privacy.

During testing, some practical limitations were noticed. Variations in lighting, camera placement, or loose clothing slightly reduced pose-detection accuracy. In addition, small fluctuations in ultrasonic sensor readings occurred when the sensor position was disturbed or obstacles were present nearby. These issues indicate that improved calibration, better sensor placement, and enhanced algorithm tuning are needed for higher reliability.

Overall, the implemented Smart Yoga Mat demonstrated dependable posture detection, quick response, and stable operation, making it a useful and affordable tool for improving yoga practice in home environments.





Fig.3: -Implementation diagram

6. DISCUSSION

The developed Smart Yoga Mat shows that combining IoT sensing with computer vision can support safe yoga training without continuous supervision. Real-time feedback increases user awareness and lowers the possibility of incorrect exercise. The system is simple to set up and relatively low-cost compared with trainer-based solutions. However, real-world usage introduces challenges such as lighting changes, sensor accuracy, and differences in body shape. Addressing these issues will require better training datasets, adaptive algorithms, and improved sensor calibration. Data-privacy protection for webcam use is also important. Even with these challenges, the system demonstrates strong potential for home fitness monitoring and guided yoga practice.

7. CONCLUSION

The Smart Yoga Mat using IoT is an intelligent fitness assistance system that improves yoga posture through real-time monitoring. The system combines a webcam, ultrasonic sensor, and laptop-based processing to detect body alignment and provide corrective

feedback. By using computer vision and machine learning techniques, it offers a low-cost and user-friendly solution for safe and effective yoga practice at home. The project demonstrates how IoT technology can enhance traditional yoga training and support digital health monitoring.

FUTURE SCOPE

Future improvements can make the Smart Yoga Mat more advanced and useful. Possible enhancements include:

- Adding more sensors like pressure sensors for balance detection
- Developing a mobile app for remote monitoring
- Supporting multiple yoga poses and automatic pose recognition
- Adding voice guidance for posture correction
- Using cloud storage for progress tracking
- Improving accuracy with advanced AI model

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